

Data preprocessing is one of the most important phases in the machine learning pipeline. The quality of the input data directly affects the accuracy and reliability of the predictive model. Real-world datasets often contain missing values, duplicate records, inconsistent data formats, outliers, and features with varying scales. If these issues are not addressed, the machine learning algorithms may produce inaccurate predictions and poor generalization.

In this flood prediction project, a systematic preprocessing workflow is performed to clean, transform, and prepare the historical meteorological dataset before training the machine learning models. The preprocessing steps ensure that the dataset is consistent, unbiased, and suitable for building an accurate flood prediction model.

Data Cleaning

The first step involves examining the dataset for any inconsistencies that may affect the learning process. Data cleaning ensures that the dataset is complete, accurate, and free from redundant information.

Handling Missing Values

Missing values are common in real-world datasets and can reduce the performance of machine learning models if not handled properly. Therefore, the dataset is carefully examined to identify any null or empty values.

Handling Outliers

Outliers are observations that differ significantly from the majority of the data. In meteorological datasets, unusually high rainfall values or abnormal weather conditions may influence the learning process and reduce model accuracy.

Handling Categorical Values

Machine learning algorithms require numerical input. Therefore, categorical features must be converted into numerical representations before model training.

Procedure

- The dataset is inspected for categorical variables.
- When categorical attributes are present, **Label Encoding** is applied to convert text labels into numerical values.
- Since the meteorological dataset used in this project contains only numerical features, **no encoding was required**.

Splitting the Dataset

To evaluate the performance of the machine learning models fairly, the dataset is divided into separate training and testing subsets.

Dataset Division

- **Training Set:** 80%
- **Testing Set:** 20%
- Prediction performance is more reliable.

Feature Scaling

$$Z = \frac{X - \mu}{\sigma}$$